ARYAN GUPTA

(734) 819-3134 | guptaaryanr@gmail.com | https://www.linkedin.com/in/guptaaryanr

EDUCATION

University of Michigan September 2019 – May 2023
Major: BSE Engineering Physics GPA: 3.25/4.0, *Cum Laude*
Minors: Electrical Engineering, Computer Science, Entrepreneurship
Selected Coursework: Multivariable & Vector Calculus; Differential Equations; Matrix Algebra; Mathematical Methods in Physics; Honors Physics I-III; Advanced Mechanics, Electricity & Magnetism, Statistical & Thermal Physics; Data Structures & Algorithms; Logic Design; Signals & Systems; Embedded Systems (Intro & Advanced); Computer Architecture; Control Systems

WORK EXPERIENCE

Technip Energies Jul 2025 – Present
Specialist Engineer I – Instrumentation & Controls

- Standards & discipline onboarding: built working familiarity with ISA/IEC instrumentation and control standards, internal Technip specifications, and EPC deliverable workflows through training programs and document reviews.
- Blue hydrogen project support: reviewed vendor instrumentation documents for consistency against P&IDs, PFDs, and Cause & Effect diagrams; flagged discrepancies and supported senior engineers in document alignment and issue resolution.
- Controls documentation: assisted with verification and update of Cause & Effect diagrams by cross-referencing legacy logic, control narratives, and P&IDs to ensure functional consistency for brownfield scope changes.
- Carbon capture project exposure: contributing to early-stage I&C updates for pump addition scope, including review of control logic narratives and incremental updates to Cause & Effect documentation based on prior designs.
- Engineering rigor: applied version control, document traceability, and change-awareness practices while working within established management-of-change and QA processes.

Technip Energies Aug 2023 – Jul 2025,
Feb 2026 – Mar 2026
Technology Developer I (Applied Physics Lead, Novel Carbon Capture R&D Team)

- Established a first-principles molecular modeling capability for internal carbon capture technology development; reproducible MD/DFT pipelines (LAMMPS, NWChem) across HPC and cloud (GPU + parallel I/O) reduced simulation wall time by ~2/3 and meaningfully lowered cloud spend (verified via internal benchmarks).
- Bridged atomistic → continuum: injected MD/DFT outputs into diffusion and CFD models with a nucleation-delay term; sensitivities to critical nucleus size improved continuum accuracy and re-prioritized model assumptions; maintained diffusion-scale codebase with new verification tests.
- Physical fidelity: evaluated force fields beyond LJ (Morse, EAM, TraPPE, SPC/E; polarizable) and benchmarked interfacial energies, diffusion rates, crystallization barriers vs bench + literature with acceptable error bounds; designed and supervised crystal growth experiments.
- Automation at scale: parameter sweeps across composition-T-P grids; continuous ingest of experimental images + tables so analyses auto-updated.
- Image analytics: Python/C++ (OpenCV) pipelines with QA segmented terabyte-scale image sets; produced UQ-backed particle-size distributions (KS/AD) replacing multi-day manual workflows with a multi-hour pipeline.
- ML integration: engineered physics-informed features; PCA/UMAP/k-means for structure discovery; RF/linear surrogates identified crystallization-rate drivers later confirmed experimentally; deployed surrogate-assisted and active-learning workflows to adaptively select simulation grids, reducing simulation count while maintaining target error bounds.
- Rigor & reproducibility: multi-level UQ (sensitivities, CIs, error propagation); authored 14 internal technical notes (two-reviewer process) and a data processing code archive forming the team's reproducibility library.
- Collaboration & leadership: Integrated atomistics with MATLAB/Fortran utilities (stress/thermowell analysis for large scale experiments); mentored teammates and onboarded a new engineer to extend the pipelines.
- Strategic impact: distilled findings into decision-ready decks supporting multi-million-dollar internal funding decisions; authored a 70-page market analysis & business plan that informed the team roadmap and funding discussions.

PREPRINTS & OPEN RESEARCH SOFTWARE

ThermoBench-Consist (v1.0) Sep 2025 – Oct 2025
Preprint: 10.5281/zenodo.17489426 *Code DOI: 10.5281/zenodo.17330440*

- Designed a CPU-only diagnostic/benchmark to vet ML equations-of-state and VLE surrogates before they are plugged into CFD/combustion/process simulators.
- Formalizes four physics checks (C1-C4): ρ-p monotonicity, $\kappa_t>0$ stability, Clapeyron slope, and speed of sound (a^2) with configurable tolerances, critical-region exclusion, and near-spinodal flagging.
- Provides reference grids for CO_2 and N_2, deterministic sampling (and seeded random grids), and guardrails (phase filtering, critical-band avoidance) to reduce false positives in fragile regimes.
- Emits one-page MD/HTML reports with plots, severity badges (info/warn/fail), and a machine-readable JSON score; runs CLI <30 s and Colab notebook <60 s on a laptop (CPU-only).

- Ships a clean adapter API (finite-difference fallbacks, unit handling via pint) and a deliberately inconsistent surrogate to illustrate failure modes and expected report signatures.
- Includes tests/CI, tiny datasets, and reproducible example outputs to enable drop-in evaluation of third-party surrogates across labs.

REB-1: Robot Energy Benchmark (v0.1) Aug 2025 – Aug 2025
Code DOI: 10.5281/zenodo.17204853
- Built a WSL-friendly CLI micro-benchmark that logs power via nvidia-smi (or a deterministic demo source) and writes tidy CSV traces with reproducible 60-second workloads.
- Provides an analysis notebook that integrates power to Wh, converts to gCO_2 with a user-set grid-intensity factor, and exports bar charts and a short demo GIF for side-by-side comparisons.
- Focuses on power/impact, deliberately complementing existing ROS tools centered on latency/throughput; enables fast A/B of autonomy workloads on commodity laptops.
- Includes schema-checked outputs, lightweight tests/CI, and turn-key examples so other groups can replicate figures exactly and contribute logs.

Assessing the Limits of Graph Neural Networks for Vapor-Liquid Equilibrium May 2025 – Sep 2025
Preprint: 10.48550/arXiv.2509.10565
- Negative-results study on cryogenic mixtures: documents phase-dependent errors and liquid-phase inconsistency that prevent trained GNN surrogates from supporting VLE solvers.
- Traces failure to derivative pathologies (e.g., density/enthalpy behaviors across phases), demonstrating why seemingly accurate pointwise fits can violate global thermodynamic identities.
- Provides a diagnostic workflow (finite-difference checks, phase-aware tests, hybrid fallback with classical property libraries) and released failure cases to help other teams avoid silent breaks.

Energy-Efficient Robotics Software (2020-2024): Systematic Literature Review Jan 2025 – Aug 2025
Preprint: 10.48550/arXiv.2508.12170 *Code DOI: 10.5281/zenodo.16907564*
- 79-study synthesis of post-2020 work on energy in autonomy stacks (planning, perception, middleware, hardware acceleration) across mobile and manipulator platforms.
- Proposes a taxonomy of energy-aware techniques (e.g., scheduling/DVFS, algorithmic refactoring, compute off-loading, task-level policy design) and a reporting checklist for fair energy claims.
- Quantifies coverage gaps (e.g., Wh/mission, energy-latency trade-offs, standardized workloads) and surfaces under-measured components (sensing, comms, OS services).
- Releases a full replication package (screening spreadsheet, code to regenerate figures, inclusion/exclusion rationale) to support future surveys and benchmark design in the community.
- Provides the analytical springboard for later software artifacts (e.g., REB-1's Wh→gCO_2 framing and quick-run workloads).

RESEARCH EXPERIENCE

BIRDS Lab, UMich Jan 2021 – Apr 2023
Research Assistant
- Built adaptable quadcopter HW/SW stack; implemented Wi-Fi ↔ UART bridges (ESP32/FTDI) for real-time control/telemetry; redesigned power boards; contributed to motor control reliability in a 5-person team.

MiTEE (Miniature Tether Electrodynamics Experiment) Jan 2020 – Dec 2020
Plasma Team; Liaison to Structures
- Studied electrodynamic tether propulsion feasibility for pico/nanosats; supported thermodynamic analysis & electronics prototyping; coordinated interfaces with Structures team.

PROJECTS

Electric Longboard Remote Controller Sep 2022 – Dec 2022
- Designed a modular remote on STM32L010C6T6 with UART/LPUART smartphone link; added speed/turn indicators, music control, battery telemetry, and display updates. Modified VESC firmware (GPIO/PWM) and produced a 3D-printed enclosure; delivered under 3.5-month / $1k constraints.

Out-of-Order Processor Architecture Feb 2022 – Apr 2022
- Implemented a MIPS R10K-style OOO core in System Verilog with I/D caches, LSQ, and branch predictor; 2-way superscalar issue and precise exceptions; added baseline SMT (round-robin), cache optimizations, and a visual debugger for bring-up.

Low-Autonomy Follower Robot Feb 2022 – Apr 2022
- Built a vision + ultrasonic follower platform (Arduino/Nucleo, PID control) using LED-encoded commands; warehouse POC for leader-follower flows.

TEACHING EXPERIENCE

UMich College of Engineering Sep 2022 – Apr 2023

EECS 314 (Circuit Design and Analysis) Instructional Aide
- Led discussion groups for ~80 students; ran lab of 20 on DC/AC circuits, RLC transients, op-amps; shaped course structure for 160+ students and supported bench instrumentation (Keysight/HPE, scopes, DMMs).

LEADERSHIP & SERVICE

AI Solutions Initiative – Americas Dec 2025 – Present
Co-Founder & Co-Lead
- Co-founded and co-lead an internal, Americas-wide AI solutions initiative to identify, develop, and pilot high-value AI and automation opportunities across EPC engineering organization.
- Established a repeatable delivery framework novel to EPC environments, incorporating agile sprints, design-verification-fitness (DVF), and Pugh decision matrices to prioritize, develop, and de-risk MVPs.
- Built and coordinated cross-disciplinary teams across multiple Americas operating centers; aligned parallel efforts to avoid duplication and enable shared tooling, datasets, and learnings.
- Partnered with engineering managers and discipline leads to compile and refine candidate project portfolios; selected and scoped initial project slate based on technical feasibility, value potential, and scalability.
- Led coordination with global digital (DigiTeam) stakeholders to ensure alignment with enterprise strategy; defined a handoff model for successful MVPs with global applicability.
- Drove internal visibility and adoption through presentations at internal events, town halls, and leadership forums; established the initiative's credibility and pipeline within months of launch.
- Currently leading development and launch of three initial AI projects with applications for instrumentation, control, piping, process, mechanical, business, and engineering methods while participating directly in hands-on technical build-out alongside team members.

Citizens for Animal Protection (CAP) Dec 2023 – Nov 2024
Volunteer
- Assisted with dog walking, kennel cleaning, and shelter operations in Houston community.

HelixCases Feb 2022 – Aug 2022
Co-founder & Software Developer
- Secured a $10k Lyft contract, designed coding competition platform reaching 550+ students at 20 universities; built UI/backend, hosted custom competitions.

StartUM Jan 2021 – Apr 2022
Member & Mentor
- Guided a pre-seed medical-volunteer startup through ideation, validation, and re-targeting of its core technology to a new market.

Project RISHI Sep 2020 – Apr 2021
Fundraising Co-Chair
- Filtered/applied to 25+ grants; secured $16k toward a $35k budget; organized campus fundraising events (raffles, trivia nights, crowdfunding, game nights) with 100+ participants.

SKILLS

- **Programming & Numerical Computing:** Python, C/C++, MATLAB/Simulink
- **Modeling, Simulation & Control:** MD/DFT (LAMMPS, NWChem), atomistic-to-continuum coupling, classical control (PID)
- **HPC, Cloud & Data Workflows:** SLURM, GPU computing, profiling/optimization, reproducible cloud & cluster pipelines
- **Computer Vision & Machine Learning:** scikit-learn (classical ML), OpenCV, PyTorch (model prototyping & experimentation)
- **Embedded & Control Systems:** STM32/ARM microcontrollers, FreeRTOS, Linux/Unix environments
- **Engineering Rigor & Reproducibility:** Git-based workflows, Docker/Conda environments, CI, technical documentation, uncertainty quantification

HONORS

- University Honors: 2019, 2020
- Dean's List: 2019, 2020